

Homework Problem

- Your company considers investing in a project with these two-year projections. What are the project's FCFs?
- Your (publicly-traded) company has an (equity) beta of 1.5, it has 40% debt and 60% equity, what is your company's expected return to equity? What is the WACC?
- Assuming the project's terminal growth rate (g) after year 2 is 3%, what is the terminal value (TV)?
- What is the NPV of the project? Should you invest?
- **Assumptions:** risk-free rate of 4%, market risk premium of 7%, corporate debt pays 6%, corporate tax rate of 25%

(Figures in \$'000)		Year 0	Year 1	Year 2
Revenue (REV)		\$0.0	\$45.0	\$75.0
Cost of Goods Sold (COGS)		\$0.0	\$10.0	\$20.0
Gross Profit (REV-COGS)		\$0.0	\$35.0	\$55.0
Operating Expenses (SG&A)		\$0.0	\$5.0	\$5.0
Depreciation (DEPR)		\$0.0	\$10.0	\$10.0
EBIT		\$0.0	\$20.0	\$40.0
Capital Expenditure (CAPX)		\$80.0	\$0.0	\$0.0
dNet Working Capital (dNWC)		\$5.0	\$5.0	\$5.0

Valuing Companies and Their Equity

DCF valuation of operating assets (OA)

- Valuing operating assets is just like valuing a project
 - Project (forecast) financials in income statement
 - Calculate free cash flows (FCFs)
 - Determine WACC
 - Find the terminal value (TV)
- Discount and sum to get present value (PV)
 - This is present value (PV) of company's operating assets (OA), i.e., the enterprise value (EV)

Balance sheet and company value

- Total company value comprises operating assets (OA) and excess cash (EC)
 - Value of operating assets (OA) is called enterprise value (EV)
 - Excess cash is cash that is free to pay out to investors
- DCF valuation of company's FCFs gives the value of the operating assets, i.e., determines enterprise value (EV)
- Value of total company: $\text{Total Value} = \text{Excess Cash} + \text{Enterprise Value} = \text{Debt} + \text{Equity}$
- Value of equity: $\text{Equity} = \text{Total Value} - \text{Debt}$
 - Often convenient to define $\text{Net Debt} = \text{Debt} - \text{Excess Cash}$
 - Then, $\text{Equity} = \text{Enterprise Value} - \text{Net Debt}$

Assets	Liabilities
Excess Cash	Debt
Operating Assets	Equity

Valuing a company share

- Valuation relation can be rearranged to value a company's equity (E)
 - **Equity = Total Value – Debt**
 - **Equity = Enterprise Value - Net Debt**
- To value an individual share, divide equity valuation by number of outstanding shares
 - If your valuation of a share exceeds the current market price, shares may be undervalued, and you should consider buying some!

Assets	Liabilities
Excess Cash	Debt
Operating Assets	Equity

Outstanding shares

- A company's shares are owned by its shareholders
 - A company can issue new shares when it needs capital (primary market)
 - For publicly-traded companies, shares are bought and sold in the stock market (secondary market)
- **“Authorized Shares”** is the maximum number of shares allowed (by corporate charter)
 - Mostly a theoretical number, not relevant for us
- **“Issued Shares”** are shares that the company has created
 - Companies can issue new shares to sell to investors to raise money for investments and projects
- **“Treasury Shares”** are shares that have been issued but are still owned by the company
- **“Outstanding Shares”** are shares that have been sold and are owned by shareholders
 - $\text{Outstanding Shares} = \text{Issued Shares} - \text{Treasury Shares}$
- To value a share of a company, divide total equity value by number of **outstanding shares**



Amazon.com Inc

✓ Added ▾

\$239.20 ↑ +42.94% (+71.86) 5Y

Jun 10, 12:41:23 PM UTC-4 · USD



Overview Earnings Financials Holdings New

📈 Amazon declined approximately 8% over the past month, pressured by a massive \$200 billion capital exp... ▾

Open	\$243.61	Volume	16.71M	Beta	1.45
High	\$250.43	P/E ratio	28.61	Shares outstanding	10.76B
Low	\$237.00	52-wk high	\$278.56	No. of employees	2M
Mkt. cap	2.58T	52-wk low	\$196.00		
Avg. vol.	44.07M	EPS	\$8.37		

Amazon

- Balance Sheet from 1Q2026 Amazon Quarterly filing

AMAZON.COM, INC.
Consolidated Balance Sheets
(in millions, except per share data)
(unaudited)

	December 31, 2025	March 31, 2026
ASSETS		
Current assets:		
Cash and cash equivalents	\$ 86,810	\$ 101,816
Marketable securities	36,219	41,273
Inventories	38,325	36,534
Accounts receivable, net and other	67,729	75,532
Total current assets	229,083	255,155
Property and equipment, net	357,025	397,458
Operating leases	86,054	88,741
Goodwill	23,273	23,449
Other assets	122,607	151,827
Total assets	\$ 818,042	\$ 916,630
LIABILITIES AND STOCKHOLDERS' EQUITY		
Current liabilities:		
Accounts payable	\$ 121,909	\$ 124,749
Accrued expenses and other	75,520	71,120
Unearned revenue	20,576	20,887
Total current liabilities	218,005	216,756
Long-term lease liabilities	87,339	90,814
Long-term debt	65,648	119,074
Other long-term liabilities	35,985	48,072
Commitments and contingencies		
Stockholders' equity:		
Preferred stock (\$0.01 par value; 500 shares authorized; no shares issued or outstanding)	—	—
Common stock (\$0.01 par value; 100,000 shares authorized; 11,246 and 11,269 shares issued; 10,731 and 10,754 shares outstanding)	112	113
Treasury stock, at cost	(7,837)	(7,837)
Additional paid-in capital	140,024	143,979
Accumulated other comprehensive income (loss)	28,230	24,868
Retained earnings	250,536	280,791
Total stockholders' equity	411,065	441,914
Total liabilities and stockholders' equity	\$ 818,042	\$ 916,630

Amazon shares

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- Amazon (AMZN) has issued 11.269B shares. Of these, 10.754B are outstanding (numbers are in millions)
 - Book value of common shares and total stockholders' equity are unrelated to market cap

Valuation with Multiples

Valuation using multiples

- In addition to DCF, companies and shares are often valued using multiples
 - Two most common multiples are: (1) price-earnings (P/E) and (2) enterprise value (EV) / EBITDA multiples
- Idea is to get multiple from comparable companies
 - Same type of business or industry averages
 - Often average multiple for number of comparables
- Apply multiple to your company's situation to get valuation

(1) Price-Earnings (P/E)

- Price-Earnings is price per share over earnings per share (EPS)
 - Earnings are net income (after taxes and interest, attributable to common shares)
- **Pros:** Easy, transparent, straightforward
- **Cons:** Does not account for leverage, risk, and future growth; can be difficult to interpret (is one company overvalued, or is the other undervalued?)
- Illustrative example:
 - AMZN is trading at P/E ratio of 28.61
 - AAPL has EPS of \$8.27
 - Using the P/E ratio from AMZN values AAPL at \$236.60 ($= \8.27×28.61) per share



Amazon.com Inc

✓ Added

\$239.20 ↑ +42.94% (+71.86) 5Y

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Mkt. cap	2.58T	52-wk low	\$196.00		
Avg. vol.	44.07M	EPS	\$8.37		

\$292.71 ↑ +129.85% (+165.36) 5Y

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Overview Earnings Financials Holdings New

📈 Apple advanced approximately 14% over the past month, fueled by record fiscal Q2 results and a massive... ▼

Open	\$290.74	Dividend	0.37%	EPS	\$8.27
High	\$300.75	Quarterly dividend	\$0.27	Beta	1.09
Low	\$287.38	Ex-dividend date	May 11, 2026	Shares outstanding	14.69B
Mkt. cap	4.30T	P/E ratio	35.39	No. of employees	166K
Avg. vol.	51.52M	52-wk high	\$317.40		
Volume	20.98M	52-wk low	\$195.07		

(2) Enterprise Value (EV) / EBITDA

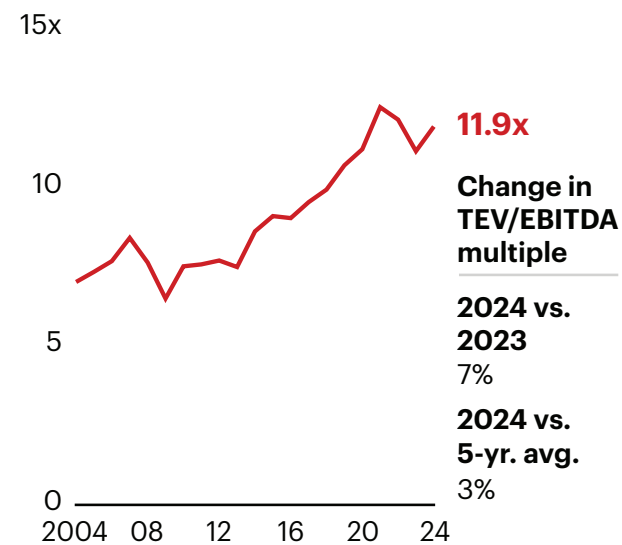
- This multiple is often used to value an entire company
 - Remember, enterprise value (EV) is value of operating assets
- EBITDA is accounting measure of core operating profitability
 - $EBITDA = EBIT + DEPR$
- Hence, EV/EBITDA values entire company relative to profitability of its operating assets
- **Example:** A typical EV/EBITDA multiple is 11.9x. A company has EBITDA of \$10m, hence enterprise value (EV) should be \$119m

Global Private Equity Report 2025

Figure 11: Deal multiples remain at or near record levels in North America and Europe

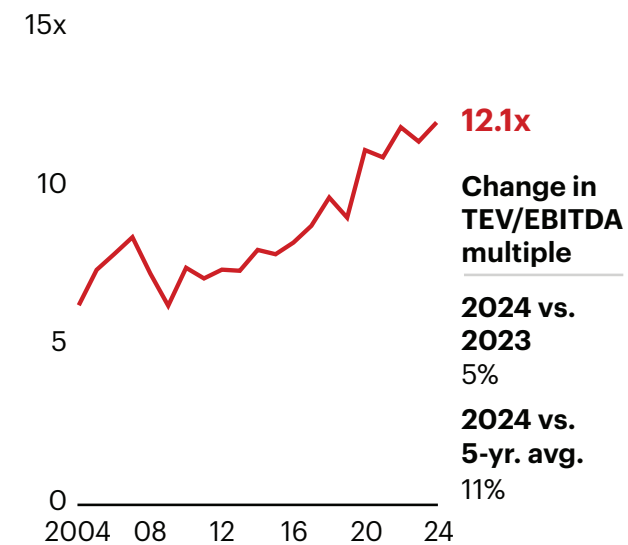
North America

Median total enterprise value (TEV)/EBITDA multiple



Western Europe

Median TEV/EBITDA multiple



Note: Data as of September 30, 2024
 Source: SPI by StepStone

- **Bain & Company:** Global Private Equity Report 2025

Using multiples for terminal value (TV)

- Terminal value (TV) is enterprise value (EV) at end of projection horizon
- When calculating TV with growing perpetuity formula, we are implicitly using an EV/FCF multiple
- **Example:** $g = 3\%$ and $WACC = 13\%$ implies 10.3x multiple
- Thus, we can use the EV/EBITDA multiple as another calculation of the TV
 - Avoids having to specify terminal growth rate (g)

$$TV = FCF \frac{(1 + g)}{WACC - g}$$

Item (figures in \$'000)		Year 0	Year 1	Year 2	Year 3	Year 4	
Revenue (REV)		\$0.0	\$400.0	\$440.0	\$480.0	...	
Cost of Goods Sold (COGS)		\$0.0	\$220.0	\$240.0	\$260.0	...	
Gross Profit (REV-COGS)		\$0.0	\$180.0	\$200.0	\$220.0	...	
Operating Expenses (SG&A)		\$0.0	\$120.0	\$120.0	\$120.0	...	
EBITDA		\$0.0	\$60.0	\$80.0	\$100.0		
Depreciation (DEPR)		\$0.0	\$40.0	\$40.0	\$40.0	...	
EBIT (Gross Profit-SG&A-DEPR)		\$0.0	\$20.0	\$40.0	\$60.0	...	
Capital Expenditure (CAPX)		\$120.0	\$0.0	\$0.0	\$0.0	...	
dNet Working Capital (dNWC)		\$15.0	\$5.0	\$5.0	\$5.0	...	
Corporate Tax Rate (T)	25%						
Free Cash Flow (FCF)		-\$135.0	\$50.0	\$65.0	\$80.0	...	
		0	1	2	3	4	
Item (figures in \$'000)		Year 0	Year 1	Year 2	Year 3	Year 4	
Free Cash Flow (FCF)		-\$135.0	\$50.0	\$65.0	\$80.0	...	
PV of FCF		-\$135.0	\$44.6	\$51.8	\$56.9		
PV		\$18.4					
Terminal Value (TV)					\$1,190.0	= EBITDA x 11.9	
growth rate (g)	3%						
WACC	12%						
PV of TV		\$847.02					
Total PV (FCF + TV)		\$865.42					

Sensitivity

Goal of company valuation

- When valuing a company, the goal is not to calculate a single “perfect” dollar value
- Rather, the goal is to develop an understanding of overall value, and value drivers and sensitivities
 - Compare valuation calculations using different methodologies and relative to other companies in the market
- Ideally, the goal is to produce a narrative story of the company’s valuation

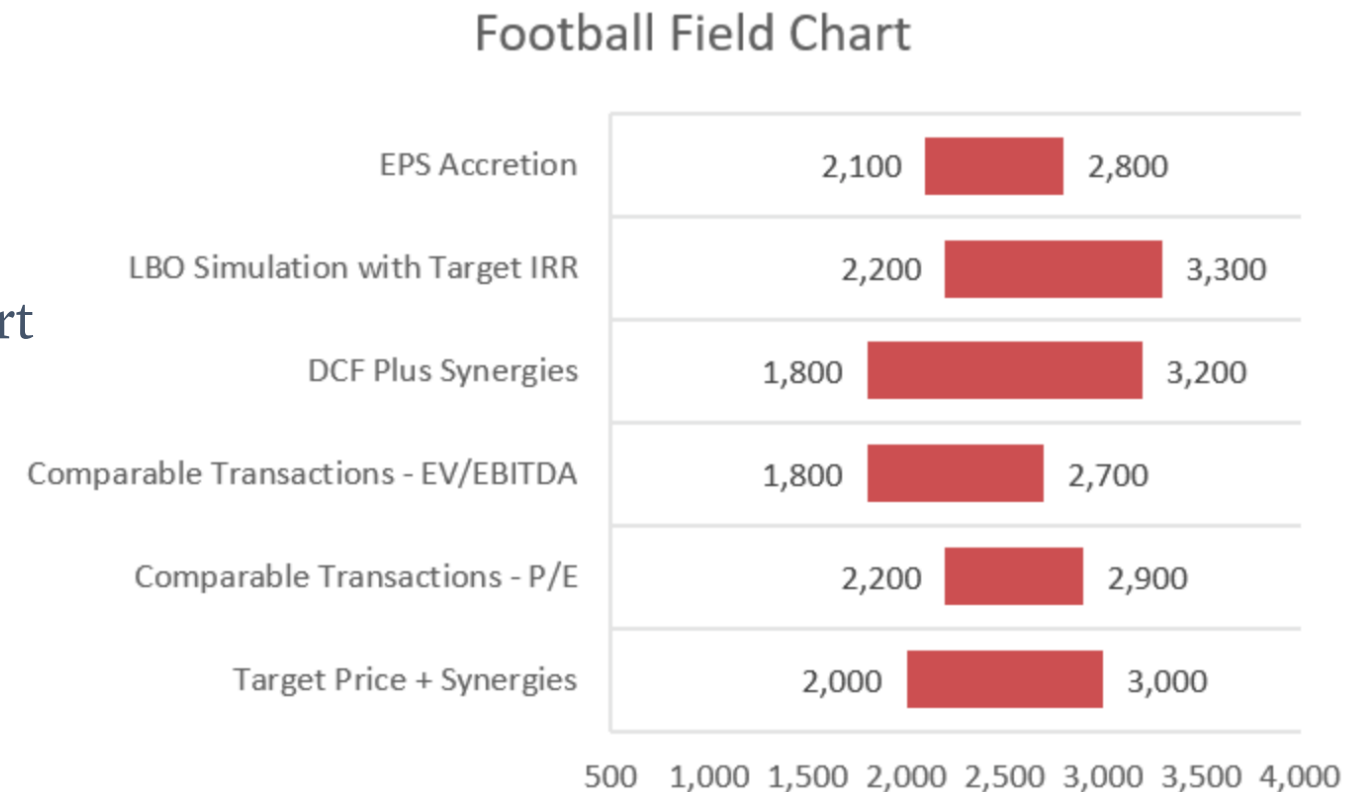
Sensitivity

Weighted average cost of capital / Terminal						
-	2.5%	3.0%	3.5%	4.0%	4.5%	5.0%
6.0%	-	-	-	-	-	-
6.5%	-	-	-	-	-	-
7.0%	-	-	-	-	-	-
7.5%	-	-	-	-	-	-
8.0%	-	-	-	-	-	-
8.5%	-	-	-	-	-	-
9.0%	-	-	-	-	-	-
9.5%	-	-	-	-	-	-
10.0%	-	-	-	-	-	-

- Last step in a valuation analysis is to explore sensitivity of results to various assumptions
- Often useful to distinguish sensitivity to financial assumptions (WACC, terminal growth rate, etc.) and operational assumptions (gross margins, , revenue growth, etc.)

Football field

- Finally, there is not a single correct valuation method or number
 - In practice, different methods and assumptions lead to different valuations
- Commonly summarized in a Football Field Chart
- Chart shows valuation ranges across methods and assumptions
- Try to reconcile different valuations across methods to get insight and confidence in valuation
 - Not trying to find a single “perfect” method
 - Instead, do all of the above and reconcile



Conclusion

This is Finance

1. Risk-Free Cash flows
2. Bond Market
3. Financial Risks (Idiosyncratic and Systematic Risk)
4. Measuring Systematic Risk (Beta)
5. Capital Asset Pricing Model (CAPM)
6. Investment Decisions (NPV)
7. Free Cash flows (FCFs)
8. Weighted Average Cost of Capital (WACC)
9. Terminal Value (TV)
10. Company Valuation and Enterprise Value (EV)

THANK YOU